

LOW-STYRENE SBR-RELATED COPOLYMERIZATION AND RUBBER COMPOSITION PATENTS

1. ABSTRACT

This report reviews two primary patents relevant to low-styrene styrene-butadiene rubber (SBR) synthesis and formulation, focusing on how styrene incorporation, microstructure, and blending strategy are controlled to tune tire-relevant properties. Although neither patent is a classic low-styrene emulsion SBR disclosure, both address composition/process design in styrene-butadiene-based elastomers where styrene content and distribution are central to performance. US10759885B2 discloses solution polymerization of branched conjugated diene/vinyl copolymers followed by optional hydrogenation, with examples using butadiene/styrene and branched dienes such as myrcene and farnesene; the patent emphasizes processability, bleeding suppression, and tire tread performance through hydrogenated branched copolymers blended into SBR-containing rubber compositions. US11549003B2 discloses a solution SSBR architecture with two parts of different vinyl content, prepared by adiabatic temperature-rise polymerization in cyclohexane using n-butyllithium, randomizing agents, and delayed monomer addition to suppress styrene block formation while maintaining high aromatic vinyl content and compatibility with BR or NR. Across the landscape, the dominant synthesis approach is solution polymerization rather than emulsion polymerization, with strong emphasis on monomer feed sequencing, randomization control, and post-polymer modification. Key formulation trends include blending with high-cis BR or natural rubber, use of silica/silane reinforcement systems, and targeting a balance of wet grip, rolling resistance, abrasion resistance, and processability. For low-styrene SBR specifically, the evidence suggests that practical routes rely less on simply lowering styrene to very low levels and more on controlling styrene distribution, block suppression, and compatibility through microstructure engineering and blending. Recommended synthesis parameters from the evidence include controlled monomer feed ratios, low block styrene formation, moderate molecular weight, and solution polymerization with initiator/randomizer systems that enable composition gradients or segmented microstructures.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10759885B2
- Patent Title: Hydrogenated branched conjugated diene copolymer, rubber composition and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of branched conjugated diene compounds with vinyl monomers using n-butyllithium in hexane/THF, followed by optional hydrogenation over palladium carbon. The resulting branched conjugated diene copolymers are then blended into rubber compositions for tires, with the examples showing both non-hydrogenated and hydrogenated variants. - Hexane: 2000 ml — Preparation Example 1 - Butadiene: 110 g — Preparation Example 1 - Styrene: 90 g — Preparation Example 1 - TMEDA: 0.22 mmol — Preparation Example 1 - n-Butyllithium: 35 mmol — Preparation Example 1 - Polymerization time: 5 hours — Preparation Example 1 - Reaction temperature: 50° C. — Preparation Example 1 - Isopropanol/hexane quench: 1.15 ml of 1 M solution — Preparation Example 1 - THF: 300 g — Preparation Example 2 - Palladium carbon: 10 g of 10% Pd/C — Preparation Example 2 - Hydrogen pressure: 5.0 kgf/cm² — Preparation Example 2 - Hydrogenation temperature: 80° C. — Preparation Example 2 - Degree of hydrogenation: 50% — Preparation Example 2 - Hexane: 500 ml — Preparation Example 3 - THF: 46 g — Preparation Example 3 - n-Butyllithium: 45 mmol — Preparation Example 3 - Isoprene: 150 g — Preparation Example 3 - Styrene: 125 g — Preparation Example 3 - Addition time: 2 hours — Preparation Example 3 - Isopropanol/hexane quench: 20 ml of 2 M solution — Preparation Example 3 - Hexane: 500 ml — Preparation Example 5 - THF: 46 g — Preparation Example 5 - n-Butyllithium: 40 mmol — Preparation Example 5 - Myrcene: 150 g — Preparation Example 5 - Styrene: 125 g — Preparation Example 5 - Addition time: 2 hours — Preparation Example 5 - Isopropanol/hexane quench: 10 ml of 2 M solution — Preparation Example 5 - Hexane: 500 ml — Preparation Example 7 - THF: 46 g — Preparation Example 7 - n-Butyllithium: 40 mmol — Preparation Example 7 - Butadiene: 95 g — Preparation Example 7 - Myrcene: 55 g — Preparation Example 7 - Styrene: 125 g — Preparation Example 7 - Addition time: 2 hours — Preparation Example 7 - Isopropanol/hexane quench: 10 ml of 2 M solution — Preparation Example 7 - Butadiene: 95 g replaced by isoprene: 95 g — Preparation Example 9 - Hexane: 2000 ml — Preparation Example 11 - Butadiene: 110 g — Preparation Example 11 - Styrene: 90 g — Preparation Example 11 - TMEDA: 0.22 mmol — Preparation Example 11 - n-Butyllithium: 35 mmol — Preparation Example 11 - Polymerization time: 5 hours — Preparation Example 11 - Reaction temperature: 50° C. — Preparation Example 11 - Isopropanol/hexane quench: 1.15 ml of 1 M solution — Preparation Example 11 - THF: 300 g — Preparation Example 12 - Palladium carbon: 10 g of 10% Pd/C — Preparation Example 12 - Hydrogen pressure: 5.0 kgf/cm² — Preparation Example 12 - Hydrogenation temperature: 80° C. — Preparation Example 12 - Degree of hydrogenation: 50% — Preparation Example 12 - Hexane: 500 ml — Preparation Example 13 - THF: 46 g — Preparation Example 13 - n-Butyllithium: 45 mmol — Preparation Example 13 -

Isoprene: 150 g — Preparation Example 13 - Styrene: 125 g — Preparation Example 13 - Addition time: 2 hours — Preparation Example 13 - Isopropanol/hexane quench: 20 ml of 2 M solution — Preparation Example 13 - Farnesene: 150 g — Preparation Example 15 - Styrene: 125 g — Preparation Example 15 - n-Butyllithium: 40 mmol — Preparation Example 15 - Hexane: 500 ml — Preparation Example 15 - THF: 46 g — Preparation Example 15 - Addition time: 2 hours — Preparation Example 15 - Isopropanol/hexane quench: 10 ml of 2 M solution — Preparation Example 15 - Farnesene: 55 g — Preparation Example 17 - Butadiene: 95 g — Preparation Example 17 - Styrene: 125 g — Preparation Example 17 - n-Butyllithium: 40 mmol — Preparation Example 17 - Hexane: 500 ml — Preparation Example 17 - THF: 46 g — Preparation Example 17 - Addition time: 2 hours — Preparation Example 17 - Isopropanol/hexane quench: 10 ml of 2 M solution — Preparation Example 17 - Butadiene: 95 g replaced by isoprene: 95 g — Preparation Example 19

Relevant Experimental Evidence - Example 1: A butadiene/styrene copolymer was prepared by solution polymerization with n-BuLi/TMEDA in hexane at 50° C., then compounded and vulcanized into a tire. - Example 2: The copolymer was hydrogenated over 10% Pd/C in THF under 5.0 kgf/cm² hydrogen at 80° C. to 50% hydrogenation. - Example 3: A styrene/isoprene copolymer was synthesized by dropwise monomer addition over 2 hours with nearly 100% polymerization. - Example 5: A styrene/myrcene copolymer was prepared under the same anionic solution conditions, showing the use of branched dienes. - Example 7: A ternary butadiene/myrcene/styrene copolymer was made, demonstrating incorporation of both linear and branched dienes before termination.

Technical Relevance This patent is relevant because it uses styrene-butadiene-based solution copolymer chemistry and explicitly addresses processability and tire performance through composition and hydrogenation control. While not a low-styrene SBR patent per se, it provides a route to tune styrene-containing elastomers and their compatibility in rubber blends.

Patent 2

- Patent Number: US11549003B2
- Patent Title: Copolymer, rubber composition, and method of preparing the same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene-butadiene copolymers in cyclohexane using n-butyllithium, a randomizing agent, and a modifier, with adiabatic temperature-rise control and delayed addition of butadiene to create a two-part copolymer having different vinyl contents. The resulting copolymer is blended with high-cis BR or natural rubber and silica/silane packages to improve compatibility

and tire properties. - Cyclohexane: 1,680 g — Preparation Example 1 - Styrene: 122.3 g — Preparation Example 1 - 1,3-Butadiene: 143 g — Preparation Example 1 - 1-(4-vinylbenzyl)pyrrolidine: 6.2 g — Preparation Example 1 - Ditetrahydrofurylpropane (DTHFP): 6.3 g — Preparation Example 1 - Piperidine: 3.2 g — Preparation Example 1 - n-Butyllithium: 6.3 g — Preparation Example 1 - Additional n-butyllithium: 9.1 g — Preparation Example 1 - Temperature rise trigger: 30° C. from initiation temperature — Preparation Example 1 - Additional 1,3-butadiene: 38.1 g — Preparation Example 1 - Addition time for additional butadiene: 5 minutes — Preparation Example 1 - Triethoxysilylpropyldiethylamine: 1.1 molar equivalents based on total monomers — Preparation Example 1 - Cyclohexane: 1,680 g — Preparation Example 2 - Styrene: 123.1 g — Preparation Example 2 - 1,3-Butadiene: 144.3 g — Preparation Example 2 - 1-(4-vinylbenzyl)pyrrolidine: 6.3 g — Preparation Example 2 - Ditetrahydrofurylpropane: 7.2 g — Preparation Example 2 - Piperidine: 3.2 g — Preparation Example 2 - n-Butyllithium: 6.4 g — Preparation Example 2 - Additional n-butyllithium: 9.2 g — Preparation Example 2 - Temperature rise trigger: 30° C. from initiation temperature — Preparation Example 2 - Additional 1,3-butadiene: 38.0 g — Preparation Example 2 - Addition time for additional butadiene: 5 minutes — Preparation Example 2 - Triethoxysilylpropyldiethylamine: 1.1 molar equivalents based on total monomers — Preparation Example 2 - Styrene content in copolymer: 40% — Table 4 / Preparation Example 1-2 - Total vinyl content in copolymer: 26% — Table 4 / Preparation Example 1-2 - Total weight average molecular weight: 180,000 g/mol — Table 4 / Preparation Example 1-2 - First block styrene content: 30% — Table 4 / Preparation Example 1-2 - First block vinyl content: 35% — Table 4 / Preparation Example 1-2 - First block weight average molecular weight: 80,000 g/mol — Table 4 / Preparation Example 1-2 - Second block styrene content: 50% — Table 4 / Preparation Example 1-2 - Second block vinyl content: 19% — Table 4 / Preparation Example 1-2 - Second block weight average molecular weight: 100,000 g/mol — Table 4 / Preparation Example 1-2 - Styrene content in copolymer: 40% — Table 4 / Preparation Example 2-2 - Total vinyl content in copolymer: 33% — Table 4 / Preparation Example 2-2 - Total weight average molecular weight: 181,000 g/mol — Table 4 / Preparation Example 2-2 - First block styrene content: 30% — Table 4 / Preparation Example 2-2 - First block vinyl content: 38% — Table 4 / Preparation Example 2-2 - First block weight average molecular weight: 90,000 g/mol — Table 4 / Preparation Example 2-2 - Second block styrene content: 50% — Table 4 / Preparation Example 2-2 - Second block vinyl content: 28% — Table 4 / Preparation Example 2-2 - Second block weight average molecular weight: 90,000 g/mol — Table 4 / Preparation Example 2-2 - Weight ratio of first part to second part: 1:0.5 to 1.5 — Claim 3 - Weight average molecular weight: 100,000 to 250,000 — Claim 4 - Molecular weight distribution: 1.0 to 1.2 — Claim 1 - Total vinyl content in copolymer: 26 to 33% by weight — Claim 1 - Aromatic vinyl-based monomer content: 38 to 42% by weight — Claim 1 - Weight ratio of aromatic vinyl to first diene to second diene: 37.5 to 42.5:45 to 50:10 to 15 — Method claim - Solvent content: 4.5 to 6.5 times total monomer weight — Method claim

Relevant Experimental Evidence - Example 1: A two-part SSBR was prepared by adiabatic temperature-rise solution polymerization with delayed butadiene addition and silane termination. - Example 2: A closely related SSBR formulation was made with slightly different monomer and randomizer amounts, yielding a higher total vinyl content. - Example 3: The patent shows that styrene content can be controlled by varying the initial aromatic vinyl/conjugated diene ratio and the timing of diene addition. - Example 4: Blending the copolymer with high-cis BR and

silica/silane gave improved compound properties versus comparative SSBRs. - Example 5: The disclosed copolymer also showed good compatibility with natural rubber in a mixed tread formulation.

Technical Relevance This patent is highly relevant because it directly addresses styrene-butadiene copolymer design, styrene distribution, and solution polymerization control for tire applications. Its segmented microstructure and delayed monomer addition strategy are useful for understanding how to manage styrene content and compatibility in low-styrene or reduced-styrene SBR-like systems.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Both patents use solution polymerization rather than emulsion polymerization, with anionic initiator chemistry centered on n-butyllithium.
- US10759885B2 focuses on branched diene/vinyl copolymers and hydrogenation, while US11549003B2 focuses on segmented SSBR microstructure with two parts of different vinyl content.
- Styrene content in US10759885B2 examples is typically 90 g styrene with 110 g butadiene or 125 g styrene with branched dienes, whereas US11549003B2 claims a higher aromatic vinyl content of 38 to 42 wt% in the copolymer.
- US10759885B2 uses TMEDA in at least one example and THF in others; US11549003B2 uses DTHFP, piperidine, and 1-(4-vinylbenzyl)pyrrolidine as randomizing/modifying agents.
- US10759885B2 includes hydrogenation at 5.0 kgf/cm² and 80° C. to 50% hydrogenation; US11549003B2 does not disclose hydrogenation but instead uses delayed monomer addition and silane coupling-agent termination.
- US11549003B2 explicitly controls block-like composition by adding 1,3-butadiene when the temperature rises 30° C. from initiation, whereas US10759885B2 emphasizes complete polymerization and optional post-hydrogenation.
- For formulation, both patents target tire compounds and use silica/silane or rubber blending strategies, but US11549003B2 more explicitly compares compatibility with high-cis BR and natural rubber.

4. CONCLUSION

The evidence indicates that low-styrene SBR development is best approached through solution-polymerization microstructure control rather than simple styrene reduction alone. US10759885B2 shows that styrene-containing copolymers can be tuned through branched diene selection and hydrogenation to improve processability and tire performance, while US11549003B2 demonstrates that segmented SSBRs with controlled vinyl gradients and delayed monomer addition can manage compatibility and performance even at relatively high styrene levels. For

low-styrene synthesis, the most transferable parameters are controlled monomer feed ratio, suppression of styrene block formation, use of randomizing agents/modifiers, and molecular-weight control in the 100,000 to 250,000 range where disclosed. The strongest evidence supports solution polymerization with n-BuLi-based initiation, staged monomer addition, and careful control of vinyl content and block distribution as the preferred route to styrene-butadiene elastomers with improved processability and balanced tire properties.

5. REFERENCES

- US10759885B2 | Hydrogenated branched conjugated diene copolymer, rubber composition and ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10759885B2/en>
- US11549003B2 | Copolymer, rubber composition, and method of preparing the same | Not disclosed | US | 2023 | <https://patents.google.com/patent/US11549003B2/en>